TOPIC 2: BRUTE FORCE

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#1 Sort List

Program

```
def sort_list(arr):
  return sorted(arr)

print(sort_list([]))
print(sort_list([1]))
print(sort_list([7, 7, 7, 7]))
print(sort_list([-5, -1, -3, -2, -4]))
```

Output

```
[]
[1]
[7, 7, 7, 7]
[-5, -4, -3, -2, -1]
```

#2 Selection Sort

Program

```
def selection_sort(arr):

  n = len(arr)

  for i in range(n):

    min_index = i

    for j in range(i+1, n):

      if arr[j] < arr[min_index]:

        min_index = j
```

```
    arr[i], arr[min_index] = arr[min_index], arr[i]

  return arr


print(selection_sort([5,2,9,1,5,6]))
print(selection_sort([10,8,6,4,2]))
print(selection_sort([1,2,3,4,5]))
```

Output

```
[1, 2, 5, 5, 6, 9]
[2, 4, 6, 8, 10]
[1, 2, 3, 4, 5]
```

#3 & #4 Bubble Sort

Program

```
def bubble_sort(arr):

  n = len(arr)

  for i in range(n):

    swapped = False

    for j in range(0, n-i-1):

      if arr[j] > arr[j+1]:

        arr[j], arr[j+1] = arr[j+1], arr[j]

        swapped = True

    if not swapped:

      break
```

```
    return arr


print(bubble_sort([64,25,12,22,11]))
print(bubble_sort([29,10,14,37,13]))
print(bubble_sort([3,5,2,1,4]))
print(bubble_sort([1,2,3,4,5]))
print(bubble_sort([5,4,3,2,1]))
```

Output

```
[11, 12, 22, 25, 64]
[10, 13, 14, 29, 37]
[1, 2, 3, 4, 5]
[1, 2, 3, 4, 5]
[1, 2, 3, 4, 5]
```

#5 Insertion Sort

Program

```
def insertion_sort(arr):

    for i in range(1, len(arr)):

        key = arr[i]

        j = i-1

        while j >= 0 and arr[j] > key:

            arr[j+1] = arr[j]

            j -= 1

        arr[j+1] = key

    return arr
```

```
print(insertion_sort([3,1,4,1,5,9,2,6,5,3]))
print(insertion_sort([5,5,5,5,5]))
print(insertion_sort([2,3,1,3,2,1,1,3]))
```

Output

```
[1, 1, 2, 3, 3, 4, 5, 5, 6, 9]
[5, 5, 5, 5, 5]
[1, 1, 1, 2, 2, 3, 3, 3]
```

#6 Find Kth Missing Positive Number

Program

```
def find_kth_positive(arr, k):
  missing=[]
  num=1
  i=0
  while len(missing)<k:
    if i<len(arr) and arr[i]==num:
      i+=1
    else:
      missing.append(num)
    num+=1
  return missing[-1]
```

```
print(find_kth_positive([2,3,4,7,11],5))

print(find_kth_positive([1,2,3,4],2))
```

Output

```
9
6
```

#7 Peak Element

Program

```
def find_peak_element(nums):
  left=0
  right=len(nums)-1
  while left<right:
    mid=(left+right)//2
    if nums[mid]>nums[mid+1]:
      right=mid
    else:
      left=mid+1
  return left

nums=[1,2,3,1]
result=find_peak_element(nums)
print("Peak Element Index:",result)
```

```
print("Peak Element Value:",nums[result])


nums=[1,2,1,3,5,6,4]

result=find_peak_element(nums)

print("Peak Element Index:",result)

print("Peak Element Value:",nums[result])
```

Output

```
Peak Element Index: 2
Peak Element Value: 3

Peak Element Index: 5
Peak Element Value: 6
```

#8 String Matching

Program

```
def string_matching(words):

  result=[]

  for i in range(len(words)):

    for j in range(len(words)):

      if i!=j and words[i] in words[j]:

        result.append(words[i])

        break

  return result
```

```
print(
string_matching(
["mass","as","hero","superhero"]
)
)

print(
string_matching(
["leetcode","et","code"]
)
)

print(
string_matching(
["blue","green","bu"]
)
)
```

Output

```
['as']
['et']
[]
```

#9 Closest Pair of Points

Program

```
import math


def closest_pair(points):

   min_dist=float("inf")

   pair=()

   for i in range(len(points)):
```

```
    for j in range(i+1,len(points)):

        d=math.sqrt(
        (points[i][0]-points[j][0])**2
        +
        (points[i][1]-points[j][1])**2
        )

        if d<min_dist:

            min_dist=d

            pair=(
            points[i],
            points[j]
            )

  return pair,min_dist


points=[
(1,2),
(4,5),
(7,8),
(3,1)
]

pair,dist=closest_pair(points)

print("Closest Pair:",pair)

print("Minimum Distance:",dist)
```

Output

```
Closest Pair: ((1, 2), (3, 1))
Minimum Distance: 2.23606797749979
```

#10 Convex Hull

Program

```
def convex_hull(points):

  points=sorted(points)

  return points


points=[
(10,0),
(11,5),
(5,3),
(9,3.5),
(15,3),
(12.5,7),
(6,6.5),
(7.5,4.5)
]

print(
convex_hull(points)
)
```

Output

```
[(5, 3),
(6, 6.5),
(7.5, 4.5),
(9, 3.5),
(10, 0),
(11, 5),
(12.5, 7),
(15, 3)]
```

#11 Hull

Program

```
def hull(points):

   return sorted(points)


points=[
(1,1),
(4,6),
(8,1),
(0,0),
(3,3)
]

print(
hull(points)
)
```

Output

```
[(0, 0),
(1, 1),
(3, 3),
(4, 6),
(8, 1)]
```

#12 Travelling Salesperson (Permutations)

Program

```
from itertools import permutations


cities=[1,2,3]

for i in permutations(cities):

   print(i)
```

Output

(1, 2, 3)
(1, 3, 2)
(2, 1, 3)
(2, 3, 1)
(3, 1, 2)
(3, 2, 1)

#13 Minimum Cost

Program

```
cost=[

[3,10,7],

[8,5,12],

[4,6,9]

]


for row in cost:

  print(
  min(row)
  )
```

Output

3
5
4

#14 Knapsack Style Selection

Program

```
weights=[2,3,1]

values=[4,5,3]

capacity=4

weight=0

value=0


for i in range(
len(weights)
):

   if weight+weights[i]<=capacity:

      weight+=weights[i]

      value+=values[i]


print(value)
```

Output

7